Inter-Animal Contact Geometry from Segmentation Masks

Do mask-derived contact features beat keypoint-hull geometry?

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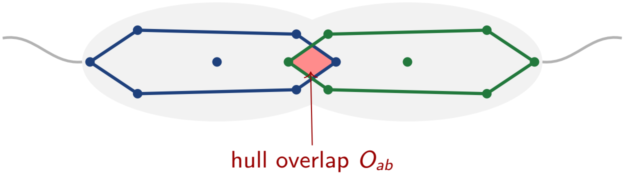
The question in one line

A **social behaviour** is defined by **how two bodies meet** — how close, how much they overlap, where they touch.

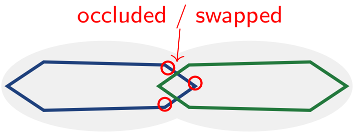
- The field reads that geometry from **keypoint hulls**.
- **We ask:** can a **segmentation mask** read it better — exactly where hulls are weakest, *during contact*?

Hull geometry (the field: SimBA, MARS)



A convex hull over each mouse's **12 keypoints**. Contact = polygon overlap $O_{ab} = |H_a \cap H_b| / |H_a \cup H_b|$ (red). Cheap — but built from **sparse points**.

Why hulls fail *during* contact

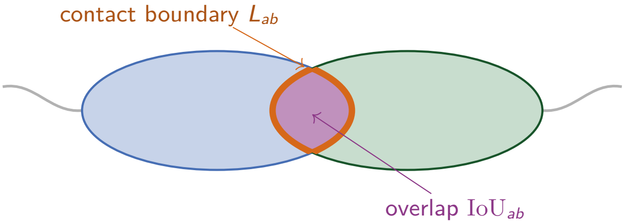


- Keypoints **occlude** one another.
- On look-alike mice the tracker **swaps** them.
- The polygon **smooths over the true boundary**.

The crux

Crudest at the very moment the behaviour is *defined*.

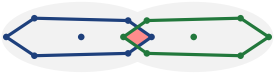
Mask geometry (our proposal, via SAM 2)



Full silhouettes from SAM 2 (same keypoints as prompts). Two numbers: overlap IoU_{ab} (plum), and **how much the outlines touch**, L_{ab} (orange).

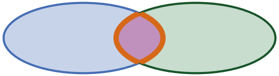
The bet: same contact, two measurements

Hull — sparse points



under-reads the contact

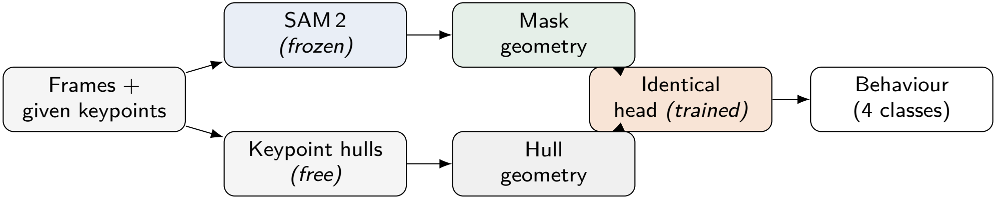
Mask — full silhouette



sees the real overlap + boundary

Same keypoints in ⇒ we compare the **representation**, not the pose. The gap should be largest on the deepest-contact behaviours.

Everything heavy is frozen and cached



SAM 2 and VideoMAE run once and are cached. The **only** trained part is a small head of **identical capacity** in every condition — so a gain means *information, not parameters*.

The data: MABe22 mouse-triplet

- **1,830** labelled 1-min clips @ 30 Hz.
- **3 look-alike mice**, **12 keypoints** each (given).
- **Per-frame labels** for four behaviours.
- Easy imaging (top-view lab); hard part is **identity through contact**.

Behaviour	**positive**
`huddles`	8.1 %
`oral_oral_contact`	0.23 %
`oral_genital_contact`	0.23 %
`chases`	0.05 %

Very rare $\Rightarrow$ report **AP**, not accuracy.

Eval (locked): four binary tasks; **split by sequence** (no leakage); per-behaviour AP + F1, macro-averaged.

The experiment (pre-registered)

- **a** **Hull geometry** — the baseline.
- **b** **Mask geometry** vs (a) — **the head-to-head**.
- **c** **Fusion** on frozen VideoMAE — does mask geometry add anything extra?

- Identical head; ≥ 5 seeds; mean $\pm$ 95 % CI.
- **Success fixed up front**; scrambled-mask control rules out a capacity artefact.

Interpretable **either way** — a clean negative is still a result.

The one make-or-break risk (settled in week 1)

Can SAM 2 keep three look-alike mice separate ***through the contact frames****?*

PASS
identities hold → proceed; contact geometry is the headline.

PARTIAL
masks merge only in deepest contact → use the *merge itself* as a high-contact feature.

FAIL
frequent swaps → pivot to **background-robustness**; report a clean negative.

A **week-1 pilot** on huddle/chase clips fires this decision. A clean spine either way.

Eight-week plan

Wk 1 Acquire + lock eval contract (done); **masking pilot** + **decision**.
Wk 2 Scoped SAM 2 mask cache (Kaggle); begin geometry features.
Wk 3 Finish contact-geometry + hull extractor; head overfit sanity.
Wk 4 **Mask vs hull head-to-head** — first real result.
Wk 5 Per-behaviour analysis; multi-seed CIs; go/no-go on MammAlps.
Wk 6 Fusion / VideoMAE-redundancy test; lock primary.
Wk 7 Background-shift robustness (insurance); consolidate.
Wk 8 Write-up; figures; package pipeline + code.

All risk front-loaded into week 1; the headline result by week 4.

Contribution

To our knowledge, the **first use of inter-animal segmentation-mask contact geometry** (mask IoU, contact-boundary length) as an explicit feature for social behaviour — in a clean head-to-head against the keypoint-hull geometry the field already uses.

- Honest: silhouette shape is *not* new; the **mask-contact measurement** and the **controlled comparison** are.
- Every heavy model frozen; reproducible from cache; multi-seed CIs.
- Built-in insurance: background-robustness if the masking pilot fails.

Thank you — questions welcome.